

Dependencies

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A key difference between a DB and a spreadsheet is that, in a DB, data is stored in multiple, interconnected tables. This raises the question of how to organize data across tables.

Let's consider our favorite pets again:

ID	name	age	food
1	Casa	9	chicken
1	Casa	9	grass
2	Kira	7	fish
2	Kira	7	grass

Here we have a new column storing their favorite food. But because each of them likes more than one kinds of food, we need two rows per cat, duplicating their name and age.

You may be tempted to replace the duplicated data with NULLs to save space:

ID	name	age	food
1	Casa	9	chicken
1	NULL	NULL	grass
2	Kira	7	fish
2	NULL	NULL	grass

But now the natural query looking for cats who like grass breaks!

```
SELECT name
  FROM pets
 WHERE food = 'grass'
```

So let's go back to duplicating the entries.

ID	name	age	food
1	Casa	9	chicken
1	Casa	9	grass
2	Kira	7	fish
2	Kira	7	grass

Can you think of a way to break up the data into multiple tables to remove the redundancy?

We can have two tables, one storing name and age, and another storing food:

ID	name	age
1	Casa	9
2	Kira	7

ID	food
1	chicken
1	grass
2	fish
2	grass

Now *name* and *age* are no longer repeated! You may notice the duplicated IDs in the second table, but that's unavoidable, and an integer takes up little space.¹

¹There's also a duplicate of *grass* which can be eliminated. Try to do that after you learn about decomposition.

Taking a step back, we see the root cause of the data redundancy is that although ID *uniquely determines* name and age, we repeat the information for every occurrence of ID:

ID	name	age	food
1	Casa	9	chicken
1	Casa	9	grass
2	Kira	7	fish
2	Kira	7	grass

Formally, there is a **functional dependency** (FD) $ID \rightarrow name$, as well as $ID \rightarrow age$.

In general, an FD $X \rightarrow Y$ (where X and Y are sets of attributes) holds for a table t , if the value of X uniquely determines the value of Y .

In our example, we also have $ID \rightarrow \text{name, age}$, but $ID \rightarrow \text{food}$ does not hold.

For an example with multiple attribute on the left, consider a table storing GPS coordinates and zipcode:

long.	lat.	zip
34N	118W	90095
47N	122W	98195

Neither the longitude nor the latitude alone uniquely determines the zipcode, but they do together, so we have long., lat. $\rightarrow$ zip

There's also some “trivial” FDs:

- ▶ For any attribute x , $x \rightarrow x$ holds
- ▶ For any sets $Y \subseteq X$, $X \rightarrow Y$ holds

We say k is a *key* if $k \rightarrow x$ holds for every attribute x in the table.² In our example, ID is a key after we break the table up, but it was not when we had one big table.

ID	name	age
1	Casa	9
2	Kira	7

ID	food
1	chicken
1	grass
2	fish
2	grass

²rigorously, a key must be *minimal*, otherwise it's a *superkey*.

“Breaking the table up” is formally known as *decomposition* or *normalization*, and the goal is to exploit FDs to remove redundancy.

In our example, we decomposed with the FD $ID \rightarrow \text{name, age}$.

In general we want to decompose so that for every FD $X \rightarrow Y$, X is a key.³⁴

³Or $X \rightarrow Y$ is *trivial*, meaning $Y \subseteq X$.

⁴More rigorously, X is a *superkey* meaning it contains a key.

But we could have decomposed in 2 steps: first with $ID \rightarrow name$, then with $ID \rightarrow age$:

ID	name
1	Casa
2	Kira

ID	age
1	9
2	7

ID	food
1	chicken
1	grass
2	fish
2	grass

This however takes up more space because the ID column is now duplicated.

Intuitively, when we break up, we want to take as much as possible, i.e., we want to decompose with an FD that has many attributes on the right.

To do this, we compute the *FD closure*: given a set of FDs Φ , the *closure* X^+ of a set X is the largest set such that $X \rightarrow X^+$ follows from Φ

Here, an FD ϕ *follows from* Φ if ϕ holds in any table t satisfying Φ .

That's all very abstract, so let's look at some cats. Suppose we have a table with these columns:

name	birth day	age	astrology sign	personality
...	...	...	...	...

Then we have:

$\text{bday} \rightarrow \text{age}$

$\text{bday} \rightarrow \text{astro}$

$\text{astro} \rightarrow \text{personality}$

We can deduce $\text{bday} \rightarrow \text{personality}$ by *composing* the FDs $\text{bday} \rightarrow \text{astro}$ and $\text{astro} \rightarrow \text{personality}$.

In general, we can deduce additional FDs using the “Armstrong axioms”:

- ▶ **Reflexivity:** $X \rightarrow Y$ if $Y \subseteq X$
- ▶ **Augmentation:** if $X \rightarrow Y$, then $XZ \rightarrow YZ$
- ▶ **Transitivity:** if $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$

We’ve seen examples of reflexivity and transitivity. Augmentation is also intuitive: from $\text{bday} \rightarrow \text{age}$ we can conclude $\text{bday, name} \rightarrow \text{age, name}$.

Actually, you don't need to remember the rules. There's a even simpler way to directly compute X^+ . The intuition can be explained with the *tableaux method*

Given the FDs $\text{bday} \rightarrow \text{age}$, $\text{bday} \rightarrow \text{astro}$, and $\text{astro} \rightarrow \text{pers.}$, let's compute bday^+ . First we write down a table with just two rows:

name	birth day	age	astrology sign	personality
Casa	3/20/2017	9	Pisces	dreamy
?	3/20/2017	?	?	?

Then, we *apply* each FD to see if we can infer the unknown values.

Step 1: Apply bday $\rightarrow$ age. Both rows share the same birthday, so the unknown age must equal 9.

name	birth day	age	astrology sign	personality
Casa	3/20/2017	9	Pisces	dreamy
?	3/20/2017	9	?	?

Step 2: Apply bday $\rightarrow$ astro. Both rows share the same birthday, so the unknown astrology sign must equal Pisces.

name	birth day	age	astrology sign	personality
Casa	3/20/2017	9	Pisces	dreamy
?	3/20/2017	9	Pisces	?

Step 3: Apply astro → pers. Both rows now share the same astrology sign, so the unknown personality must equal dreamy.

name	birth day	age	astrology sign	personality
Casa	3/20/2017	9	Pisces	dreamy
?	3/20/2017	9	Pisces	dreamy

No more FDs can fill in new values — we've reached a *fixpoint*.
The final result is:

name	birth day	age	astrology sign	personality
Casa	3/20/2017	9	Pisces	dreamy
?	3/20/2017	9	Pisces	dreamy

This tell us the closure $\text{bday}^+ = \{\text{bday}, \text{age}, \text{astro.}, \text{pers.}\}$, i.e.,
birthday uniquely determines all of these attributes.

More abstractly, the algorithm works like this:

```
def clos(FDs, V):  
    C = V  
    while C changes:  
        for (X, Y) in FDs:  
            if C contains X:  
                add Y to C  
    return C
```

In words: we start with the set V , then keep applying each FD to grow the set, until doing so doesn't change it anymore.

We can use the closure algorithm to check if a given FD $\varphi = X \rightarrow Y$ follows from a given set of FDs Φ , denoted $\Phi \vdash \varphi$:

1. Compute the closure X^+ of X over Φ
2. Check if $Y \subseteq X^+$
3. If so, $\Phi \vdash \varphi$, otherwise $\Phi \not\vdash \varphi$

That's a lot of new concepts! Let's step back and think about our motivation.

Our original table had *redundancy*, which we now understand is caused by *functional dependencies*:

ID	name	age	food
1	Casa	9	chicken
1	Casa	9	grass
2	Kira	7	fish
2	Kira	7	grass

In particular, we have FDs like $ID \rightarrow name$ where the left-hand-side ID is *not* a key.

To fix the redundancy, we break the table up according to the FDs:

ID	name	age
1	Casa	9
2	Kira	7

ID	food
1	chicken
1	grass
2	fish
2	grass

We also want to “take as much as possible” when breaking up, so we want to compute the *closure* which tells us *all* the dependent attributes.

So our algorithm so far looks like this:

- ▶ Find any FD $X \rightarrow Y$ such that $Y \not\subseteq X$ and X does not contain a key
- ▶ Compute the closure X^+ of X
- ▶ Move X^+ to a new table while keeping X in the old one

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The idea is, once we have a composite key K , we can always keep adding stuff to K to get another superkey, but K is more interesting because it doesn't contain “junk”.

⁵superset of a key

Let's practice this on the larger table as well:

name	birth day	age	astrology sign	personality
...	...	...	...	...

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name	birth day
...	...

birth day	age	astrology sign	personality
...	...	...	...

But we're not done! In the second table, we still have redundancy:

birth day	age	astrology sign	personality
3/20/2017	9	Pisces	dreamy
2/19/2016	10	Pisces	dreamy

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1. astro $\rightarrow$ pers. yet astro is not a key
2. Compute $\text{astro}^+ = \{\text{astro}, \text{pers.}\}$
3. *Factor out* $\{\text{astro}, \text{pers.}\}$

We end up with 3 tables that cannot be decomposed further:

name	birth day
...	...

birth day	age	astrology sign
...	...	...

astrology sign	personality
...	...

The general decomposition algorithm is also a *fixpoint* algorithm:

```
def decompose(R):  
    S = attributes of R  
    find nontrivial  $X \rightarrow Y$  s.t.  $X$  contains no key  
    if not found:  
        return  
    else:  
        C = clos(X)  
        break R into  $R_1(C)$ ,  $R_2(S - (C - X))$   
        decompose( $R_1$ )  
        decompose( $R_2$ )
```

At the end of this process, the tables are in *Boyce-Codd Normal Form*, meaning that any FD $X \rightarrow Y$ that still hold in any table will satisfy either:

- ▶ $Y \subseteq X$ (“trivial” FD)
- ▶ Some $K \subseteq X$ is a key (X is a *superkey*)

So far, we've been focusing on redundancy caused by functional dependencies, which can be addressed by decomposing according to the FDs. However, this can at most save use linear amount of space (why?). Let's now consider a more severe form of redundancy caused by *independence*.

Suppose we have a single table storing the favorite toy and food for each cat:

name	toy	food
casa	ball	chicken
casa	bag	chicken
casa	TP	chicken
casa	ball	grass
casa	bag	grass
casa	TP	grass

The issue here is that **toy** is independent of **food** for each cat, yet we store them in the same table.

The solution is once again breaking the table up:

name	toy
casa	ball
casa	bag
casa	TP

name	food
casa	chicken
casa	grass

Formally, the redundancy can be described with (conditional) independence. For that, let's review some basic probability theory.

A *distribution* assigns a probability to each event. If we draw a row uniformly at random from our data, there's $1/2$ chance we get a row with food = chicken or food = grass, So the distribution over foo is:

food	p
chicken	$1/2$
grass	$1/2$

Similarly, the distribution over toy is:

toy	p
ball	$1/3$
bag	$1/3$
TP	$1/3$

A *joint distribution* describes the probabilities of different types of events occurring at the same time. If we draw a random row, the probability of $\text{toy} = \text{ball} \wedge \text{food} = \text{chicken}$ is $1/6$, and so is for any other pairing of toy and food:

name	toy	food	p
casa	ball	chicken	$1/6$
casa	bag	chicken	$1/6$
casa	TP	chicken	$1/6$
casa	ball	grass	$1/6$
casa	bag	grass	$1/6$
casa	TP	grass	$1/6$

The *conditional* distribution focuses a “slice” of the larger distribution. For example, $P(\text{toy}|\text{food} = \text{chicken})$ is:

toy	food	p
ball	chicken	1/3
bag	chicken	1/3
TP	chicken	1/3

Two things happened to obtain this distribution:

1. We are focusing only on the rows where food =
2. We *re-normalized* the probabilities so that they sum to 1

Two types of events are *independent* if all conditional distributions agree with the unconditional ones. Formally,

$\forall b : P(A|B = b) = P(A)$ In the example above,

$P(\text{toy}) = P(\text{toy}|\text{food} = \text{chicken})$, so knowing the cat likes chicken doesn't tell us anything about the toy preference, and this also needs to hold for $\text{food} = \text{grass}$ to establish the independence.

Another way to check independence is to see if the joint distribution is simply a product of the single ones⁶:

$$P(\text{toy}, \text{food}) = P(\text{toy}) \times P(\text{food})$$

⁶A.k.a. marginal distributions.

If you are traumatized by probabilities, here's a different way to understand independence. To check if toy is independent from food, first GROUP BY one of them, say food:

food	toy
chicken	ball
chicken	bag
chicken	TP

food	toy
grass	ball
grass	bag
grass	TP

We see that across different food groups, the set of different values for toy remains the same. This tells us that the toy preference is not affected by the food preference.

You can also see that the large table is the *cartesian product* of the small ones:

name	toy	food	p
casa	ball	chicken	1/6
casa	bag	chicken	1/6
casa	TP	chicken	1/6
casa	ball	grass	1/6
casa	bag	grass	1/6
casa	TP	grass	1/6

food	toy
chicken	ball
chicken	bag
chicken	TP

food	toy
grass	ball
grass	bag
grass	TP

This is why redundancy caused by independence costs a lot more space than dependencies.

But we're not done yet. I've been having some relationship issues with my cats lately, so I gathered some data:

love_c	love_k
5	4
5	5
4	5
4	4
1	2
1	1
2	1
2	2

It looks like Kira's love towards me is dependent upon Casa's love!
How can this be?

It turns out they love me more when they are hungry:

hungery?	love _c	love _k
y	5	4
y	5	5
y	4	5
y	4	4
n	1	2
n	1	1
n	2	1
n	2	2

In statistics, hungry is called a *confounding factor*.

Now, if we *condition* on hunger, we will see the cats become independent again:

hungery?	love _c	love _k
y	5	4
y	5	5
y	4	5
y	4	4

hungery?	love _c	love _k
n	1	2
n	1	1
n	2	1
n	2	2

Formally, we say two types of events X, Y are *conditionally independent* on another type Z , if they are independent for every possibility of Z , written $X \perp Y \mid Z$.

When we have conditional independence in a table, we can break it up along the column conditioned on:

hungery?	love _c
y	5
y	4
n	1
n	2

hungery?	love _k
y	5
y	4
n	1
n	2

But we need to be careful! Suppose we have another column that somewhat depends on the rest:

hungery?	love _c	love _k	happy _r
y	5	4	2
y	5	5	1
y	4	5	2
y	4	4	3
n	1	2	2
n	1	1	1
n	2	1	2
n	2	2	3

This shows I'm happy if my cats love/hate me a little bit, but not too much.

Suppose we decompose and take out love_k :

hungery?	love_c	happy_r
y	5	?
y	4	?
n	1	?
n	2	?

Now since we have two rows with $\text{love}_c = 5$ but different happy_r , we can't collapse them any more.

But now we're in trouble: if we join these tables back together, we don't get back the original one!

hungery?	love _c	happy _r
y	5	2
y	5	1
y	4	2
y	4	3
n	1	2
n	1	1
n	2	2
n	2	3

hungery?	love _k
y	5
y	4
n	1
n	2

The reason is that, although love_c and love_k are conditionally independent, happy_r depends on both of them in a nontrivial way, and to represent this dependency faithfully, we *have* to put them in the same table.

Another example is to think about chemical experiments:

alt.	temp.	pressure	color
...	...	...	...

Although temperature and pressure are independent for each altitude, the color of your material depends on both, and you certainly don't want to decouple the pressure data in your records!